



Engineering College

An **AUTONOMOUS** Institution

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Major Project

ABSTRACT

Autonomous Code Review Agent

Software development requires regular code reviews to identify bugs, security vulnerabilities, and coding issues before deployment. However, manual code review is time-consuming, prone to human error, and depends on the developer's experience. Existing tools detect basic coding errors but often fail to provide intelligent explanations and effective fix suggestions. As software applications become more complex, developers need an automated solution that can improve code quality while reducing review time. An efficient code review process also helps maintain coding standards and improves collaboration among development teams.

The proposed **Autonomous Code Review Agent** is an AI-powered system that automatically analyzes source code, detects bugs, security vulnerabilities, and code quality issues, and recommends optimized fixes based on coding best practices. The system can integrate with GitHub repositories to review code changes, generate detailed reports, and provide clear explanations for each detected issue. It also helps developers understand coding mistakes, improve software performance, and ensure maintainable and secure applications. This project aims to reduce development effort, increase productivity, and support developers in building reliable, high-quality software through intelligent automation.

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